

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology (M.Tech.)

Sem - II University Examination, January- 2011

M.Tech. (CAD/CAM)

ME – 718 Computational Fluid Dynamics

Date: 05.01.2011, Wednesday

Time: 10:30 a.m. to 01:30 p.m.

Maximum Marks: 70

Instructions:

1. Figures to the right indicate **full** marks.
2. Make suitable assumptions and draw neat figures wherever if required.

SECTION- I

- Q1 (a)** Experimental testing stage can be reduced, but cannot be completely eliminated using CFD. Comment. **05**
- (b)** State and derive the relation for Reynolds's Transport theorem. **05**
- (c)** Enlist some important area of application of CFD. **02**

- Q2 (a)** The differential Equations of velocity potential for a study two dimensional flow is **10**
- $$\left(1 - \frac{\phi_x^2}{a^2}\right)\phi_{xx} - \frac{2\phi_x\phi_y}{a^2}\phi_{xy} + \left(1 - \frac{\phi_y^2}{a^2}\right)\phi_{yy} = 0$$
- Where ϕ is velocity potential of absolute velocity vector V. Explain the mathematical behavior of this equation.

- (b)** List down the commercially available CFD packages available in the market. **02**

OR

- Q2 (a)** What are Eigen Value problems? What are Eigen values & Eigen vectors. Compute highest Eigen value and corresponding Eigen vectors for the system below **10**
- $$3x - y = 0, -2x + 4y - 3z = 0, -y + z = 0$$
- (b)** Explain what is meant by truncation error. **02**

- Q3 (a)** Explain the Various Convection Schemes. **08**
- (b)** Explain the elliptic grid generation. **03**

OR

- Q3 (a)** Explain the approximation of Surface Integral and Volume Integral in FVM. **06**
- (b)** Explain the various Discretization techniques. **05**

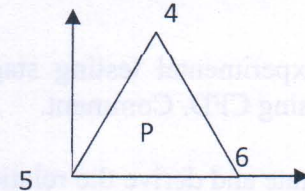
P.T.O

SECTION - II

Q4 (a) What is meant by element equation in FEM? Why is FEM termed as “piecewise approximation”? 06

(b) The nodal values of all elements following solution are as below. Compute the value of variable at a point P within the element as shown in figure. 06

Node No.	X- Coordinate	Y-Coordinate	Variable value
4	3.0	6.0	154.0
5	0.0	0.0	454.0
6	6.0	0.0	MM ²
P	(3*MM ²)/500	MM ² /500	?



Where MM is your Birth month.

Q5 (a) Explain the Crank-Nicholson form of difference Equation and its importance. 08

(b) Using Taylor's series derive second order central-difference for $\frac{\partial u}{\partial y}$. 04

OR

Q5 Discuss finite difference formulation for a plate of unit size with two of its edges subjected to steady state temperatures and other two edges insulated. 12

Q6 (a) Explain structured grid and Unstructured grid 06

(b) Draw a flow chart for the Gauss Elimination Method 05